

Habib University
Digital Logic and Design
EE/CE-172/222
Assignment 1 Solution

(d) $(1.052)_6$

2 pts

$$\begin{array}{cccc} 1 & . & 0 & 5 & 2 \\ \times & & \times & \times & \times \\ 6^0 & & 6^{-1} & + 6^{-2} & + 6^{-3} \end{array}$$
$$= (1) \cdot (0 + 0.138 + 0.009)$$
$$= (1.147)_{10}$$

Question: 02

(a) $(57.6)_{10} \rightarrow ()_2$

2 pts

Whole part:

2	57
2	28 - 1
2	14 - 0
2	7 - 0
2	3 - 1
	1 - 1

(111001)

Decimal Part:

$$\begin{array}{l} 0.6 \times 2 = 1.2 \rightarrow 1 \\ 0.2 \times 2 = 0.4 \rightarrow 0 \\ 0.4 \times 2 = 0.8 \rightarrow 0 \\ 0.8 \times 2 = 1.6 \rightarrow 1 \end{array}$$

(.1001)

Ans $\rightarrow (111001.1001)_2$

(b) $(2.31)_5 \rightarrow ()_8$ 2 pts

Convert to decimal and then octal

$(2.31)_5 \rightarrow ()_{10}$:

$$\begin{array}{r} 2 \quad . \quad 3 \quad 1 \\ \times \quad 5^0 \quad 5^{-1} \quad 5^{-2} \\ \hline (2) \quad . \quad (0.6 + 0.04) = (2.64)_{10} \end{array}$$

$(2.64)_{10} \rightarrow ()_8$:

Whole part:

$$\begin{array}{r} 8 \overline{) 26} \\ 2 \times 8 = 16 \\ \underline{10} \\ 10 - 8 = 2 \\ \hline (2) \end{array}$$

Decimal Part:

$$\begin{array}{l} 0.64 \times 8 = 5.12 \rightarrow 5 \\ 0.12 \times 8 = 0.96 \rightarrow 0 \\ 0.96 \times 2 = 1.92 \rightarrow 1 \\ \hline (.507) \end{array}$$

Ans $\Rightarrow (2.507)_8$

(c) $(145.7)_8 \rightarrow ()_{16}$ 2 pts

$(145.7)_8 \rightarrow ()_{16}$ direct conversion using binary method

$$\begin{array}{ccccccc} 1 & 4 & 5 & . & 7 & & \\ 001 & 100 & 101 & & 111 & 0 & 11 \end{array}$$

$$(0110 \ 0101 \ . \ 1110)_2$$

$$\downarrow \quad \downarrow \quad \downarrow$$

Ans $\Rightarrow (65.E)_{16}$

d) $(1010.101)_2 \rightarrow ()_7$

2 pts

To base 10:

$$\begin{array}{ccccccc}
 1 & 0 & 1 & 0 & . & 1 & 0 & 1 \\
 \times & \times & \times & \times & & \times & \times & \times \\
 2^3 & + 2^2 & + 2^1 & + 2^0 & & 2^{-1} & + 2^{-2} & + 2^{-3}
 \end{array}$$

$$= (8 + 0 + 2 + 0) + (0.5 + 0 + 0.125)$$

$$= (10.625)_{10}$$

To base 7:

Whole part:

$$\begin{array}{r}
 7 \overline{) 10} \\
 7 \times 1 = 7 \\
 \hline
 3
 \end{array}$$

$(13)_7$

Decimal part:

$$\begin{array}{r}
 0.625 \times 7 = 4.375 \rightarrow 4 \\
 0.375 \times 7 = 2.625 \rightarrow 2
 \end{array}$$

$(.42)_7$

Ans $\rightarrow (13.42)_7$

Question: 03

a) Unsigned Number

2 pts

$$\begin{array}{cccccccc}
 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0
 \end{array}$$

$$\begin{array}{cccccccc}
 \times & \times & \times & \times & \times & \times & \times & \times & \times & \times & \times
 \end{array}$$

$$2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$= (28 + 64 + 0 + 16 + 8 + 0 + 2 + 0) = (118)_{10}$$

b) Signed (2's complement)

2pts

- MSB represents sign

- Since MSB is 1, number is negative

Computing magnitude:

• 1's complement $\rightarrow 00100101$

• 2's complement $\rightarrow 00100110$

• Converting to decimal:

1 0 0 1 1 0

$\times \times \times \times \times \times$

$2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

$= (32 + 16 + 0 + 4 + 2 + 0) = 38$

Since sign was negative, we have the number $(-38)_{10}$

c) Signed 1's complement

2pts

• MSB 1 $\rightarrow$ negative

• 1's complement computed above

• Converting 1's complement to decimal for magnitude:

1 0 0 1 0 1

$\times \times \times \times \times \times$

$2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

$= (32 + 0 + 0 + 4 + 0 + 1) = (-37)_{10}$

d) BCD

1101 1010

↓

Converting to decimal we get 13, which is not a valid decimal number.

Invalid

Question: 04

(a)

4 pts

Since we have 5 unique characters, this is like a base-5 coded number system.

Actual	0	1	2	3	4	5	6	7	8	9	10	11
Base 5	0	1	2	3	4	10	11	12	13	14	20	21
Coded	(	!	#	@	!	!	!	!	!	!	#	!

(b)

2 pts

5 | 429
5 | 85 - 4 → !
17 - 0 → (
3 - 2 → #
@
↓
@

Answer ⇒ (@#!)

(c)

2 pts

##! → (224)₅

$$= 5^2 + 5^1 + 5^0 = (50 + 10 + 4) = (64)_{10}$$

Question: 05

6pts

We can find out the arbitrary base (b) by converting the equation to decimal:

$$\begin{array}{ccccccc} (44 & + & 3) & * & 2 & = & 130 \\ \overset{x}{(b^1+b^0)} & + & \overset{x}{b^0} & * & \overset{x}{b^0} & & \overset{x}{b^2} + \overset{x}{b^1} + \overset{x}{b^0} \end{array}$$

$$\rightarrow (4b+4 + 3) * 2 = b^2 + 3b$$

$$\rightarrow 8b + 14 = b^2 + 3b$$

$$\rightarrow b^2 + 3b - 8b - 14 = 0$$

$$\rightarrow b^2 - 5b - 14 = 0$$

By solving equation we get $b = 7$ and $b = -2$

Since base cannot be negative, the number system has base-7

Question: 6.

Considering an arbitrary id: 09214

Method: 1

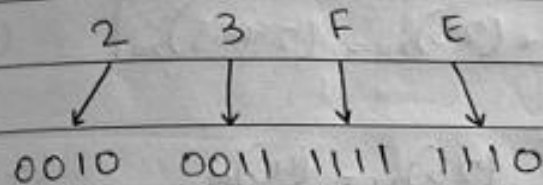
2	9214	(1 point)
2	4607 $\rightarrow$ 0	
2	2303 $\rightarrow$ 1	
2	1151 $\rightarrow$ 1	
2	575 $\rightarrow$ 1	
2	287 $\rightarrow$ 1	
2	143 $\rightarrow$ 1	
2	71 $\rightarrow$ 1	
2	35 $\rightarrow$ 1	
2	17 $\rightarrow$ 1	
2	8 $\rightarrow$ 1	
2	4 $\rightarrow$ 0	
2	2 $\rightarrow$ 0	
	1 $\rightarrow$ 0	

$$(9214)_{10} = (1000111111110)_2 \quad (1 \text{ point})$$

Method: 2

16	9214	(0.5 point)
16	575 $\rightarrow$ 14	
16	35 $\rightarrow$ 15	
	2 $\rightarrow$ 3	

$$(9214)_{10} = (23FE)_{16} \quad (0.5 \text{ point})$$



$$(23FE)_{16} = (1000111111110)_2 \quad (0.75 \text{ point})$$

∴ Second method is faster and there are less chances of error. (0.25 point)

Question: 7

a.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	1	2	3	4	5	6	7	8	9	A	B	C	D	E

15	16	17	18	19	20
F	G	H	I	J	10

(1 point)

b. $(J9)_{20} = (19 \times 20) + (9 \times 20^0) \quad (3 \text{ points})$

$$(J9)_{20} = 380 + 9$$

$$(J9)_{20} = (389)_{10}$$

$$(2B)_{20} = (2 \times 20) + (11 \times 20^0)$$

$$(2B)_{20} = 40 + 11$$

$$(2B)_{20} = (51)_{10}$$

3 point

Calculating the total charge.

$$389 \times 51 = 19839.$$

$$(J9)_{20} \times (28)_{20} = (298J)_{20} \quad (1 \text{ point})$$

c- $(F)_{20} = (15)_{10}$; $(23)_{20} = (43)_{10}$. (3 points)

Total charge after selling $(F)_{20}$ Sandwiches,

$$389 \times 15 = (5835)_{10}$$

Total cost in making $(F)_{20}$ Sandwiches,

$$43 \times 15 = (645)_{10}$$

Profit altogether,

$$5835 - 645 = 5190$$

$$(5190)_{10} = (CJA)_{20} \quad (1 \text{ point})$$

Profit per sandwich.

$$389 - 43 = 346.$$

$$(346)_{10} = (HG)_{20} \quad (1 \text{ point})$$

Question:8.

Taking name: Ali Khan.

$$A \rightarrow 50$$

$$L \rightarrow 5B$$

$$I \rightarrow 58$$

$$K \rightarrow 5A$$

$$H \rightarrow 57$$

$$A \rightarrow 50$$

$$X_1 = 505B58$$

$$X_2 = 5A5750.$$

a- Taking 16's complement of X_2 .

$$\begin{array}{r}
 15 \quad 15 \quad 15 \quad 15 \quad 15 \quad 15 \\
 - \quad 5 \quad A \quad 5 \quad 7 \quad 5 \quad 0 \\
 \hline
 A \quad 5 \quad A \quad 8 \quad A \quad F \quad (1 \text{ point})
 \end{array}$$

Adding 1.

$$\begin{array}{r}
 A \quad 5 \quad A \quad 8 \quad A \quad F \\
 + 1 \\
 \hline
 A \quad 5 \quad A \quad 8 \quad B \quad 0 \quad (0.5 \text{ point})
 \end{array}$$

Adding X_1 and 16's complement of X_2 .

$$\begin{array}{r}
 5 \quad 0 \quad 5 \quad B \quad 5 \quad 8 \\
 + A \quad 5 \quad A \quad 8 \quad B \quad 0 \\
 \hline
 F \quad 6 \quad 0 \quad 4 \quad 0 \quad 8 \quad (0.5 \text{ point})
 \end{array}$$

No overflow $\rightarrow$ result is negative.

b- Taking 15's complement of X_1 .

$$\begin{array}{r}
 15 \quad 15 \quad 15 \quad 15 \quad 15 \quad 15 \\
 - \quad 5 \quad 0 \quad 5 \quad B \quad 5 \quad 8 \\
 \hline
 A \quad F \quad A \quad 4 \quad A \quad 7 \quad (1 \text{ point})
 \end{array}$$

Adding X_2 and 15's complement of X_2 .

$$\begin{array}{r}
 5 \quad A \quad 5 \quad 7 \quad 5 \quad 0 \\
 + A \quad F \quad A \quad 4 \quad A \quad 7 \\
 \hline
 1 \quad 0 \quad 9 \quad F \quad B \quad F \quad 7 \quad (0.5 \text{ point})
 \end{array}$$

overflow

Since there is an overflow, it will be added to the sum.

0 9 F B F 7
+ 1

0 9 F B F 8 (0.5 point)

e- Converting X_1 ,

5 0 5 B 5 8
0101 0000 0101 1011 0101 1000
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
2 4 0 5 5 5 3 0

$(010100000101101101011000)_2 = (24055530)_8$
 $(505B58)_{16} = (24055530)_8$ (1.5 point)

Converting X_2 ,

5 A 5 7 5 0
0101 1010 0101 0111 0101 0000
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
2 6 4 5 3 5 2 0

$(01011010010101101010000)_2 = (26453520)_8$
 $(5A5750)_{16} = (26453520)_8$ (1.5 point)

Adding $X_1 + X_2$.

$$\begin{array}{r} 2\ 4\ 0\ 5\ 5\ 5\ 3\ 0 \\ +\ 2\ 6\ 4\ 5\ 3\ 5\ 2\ 0 \\ \hline 5\ 2\ 5\ 3\ 1\ 2\ 5\ 0 \end{array} \quad (1\ \text{point})$$

$$(24055530)_8 + (26453520)_8 = (52531250)_8.$$

Question: 9

a- $(450623)_8 - (256752)_8$.

Taking 7's complement

$$\begin{array}{r} 7\ 7\ 7\ 7\ 7\ 7 \\ -\ 2\ 5\ 6\ 7\ 5\ 2 \\ \hline 5\ 2\ 1\ 0\ 2\ 5 \end{array} \quad (1\ \text{point})$$

Adding 1.

$$\begin{array}{r} 5\ 2\ 1\ 0\ 2\ 5 \\ +\ 1 \\ \hline 5\ 2\ 1\ 0\ 2\ 6 \end{array} \quad (1\ \text{point})$$

$$\begin{array}{r} 4\ 5\ 0\ 6\ 2\ 3 \\ +\ 5\ 2\ 1\ 0\ 2\ 6 \\ \hline \end{array}$$

overflow $\rightarrow$ (1) 0 7 1 6 5 0 (1 point)
discarded

$$(450623)_8 - (256752)_8 = (71650)_8.$$

b- $(1101.001)_2 - (1011.001)_2$

Taking 1's complement.

$$\begin{array}{r} 1101.111 \\ - 1011.001 \\ \hline 0100.110 \end{array} \quad (1 \text{ point})$$

2's complement $\rightarrow 0100.111 \quad (1 \text{ point})$

$$\begin{array}{r} 1101.001 \\ + 0100.111 \\ \hline 10010.000 \end{array}$$

overflow discard $\rightarrow 0010.000 \quad (1 \text{ point})$

$(1101.001)_2 - (1011.001)_2 = (0010.000)_2$

Question: 10

a- $\begin{array}{ccc|ccc} 1 & 2 & 3 & 4 & 5 & 6 \\ 0001 & 0010 & 0011 & 0100 & 0101 & 0110 \end{array} \quad (2 \text{ point})$

$$\begin{array}{r} 0001 \quad 0010 \quad 0011 \\ + 0100 \quad 0101 \quad 0110 \\ \hline 0101 \quad 0111 \quad 1001 \end{array} \quad (1 \text{ point})$$

5 7 9

$123 + 456 = 579$

b- $523 - 239$

Taking 10's complement.

$$\begin{array}{r} 9 \quad 9 \quad 9 \\ - 2 \quad 3 \quad 9 \\ \hline 7 \quad 6 \quad 0 \end{array}$$

$$\begin{array}{r} 7 \quad 6 \quad 0 \\ + 1 \\ \hline 7 \quad 6 \quad 1 \end{array}$$

(0.5 point)

$$\begin{array}{ccc} 7 & 6 & 1 \\ \downarrow & \downarrow & \downarrow \end{array} \quad \left. \begin{array}{ccc} 5 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \end{array} \right\}$$

(0.5 point)

$$\begin{array}{ccc} 0111 & 0110 & 0001 \end{array} \quad \left. \begin{array}{ccc} 0101 & 0010 & 0011 \end{array} \right\}$$

$$\begin{array}{ccc} 0101 & 0010 & 0011 \\ + 0111 & 0110 & 0001 \\ \hline 1100 & 1000 & 0100 \end{array}$$

greater than 2

$$\begin{array}{ccc} 1100 & 1000 & 0100 \\ + 0110 & & \\ \hline 10010 & 1000 & 0100 \end{array}$$

$$\begin{array}{ccc} 10010 & 1000 & 0100 \\ 1112 & 1108 & 11014 \end{array} \quad (1 \text{ point})$$

overflow discard

$$\begin{array}{ccc} 10010 & 1000 & 0100 \\ 1112 & 1108 & 11014 \end{array}$$

$$523 - 239 = 284$$

c- $28 + 6.12$

$$\begin{array}{cc} 2 & 8 \end{array}$$

$$\begin{array}{cc} 0010 & 1000 \end{array}$$

$$\left. \begin{array}{ccc} 6 & 1 & 2 \\ 0110 & 0001 & 0010 \end{array} \right\} (1 \text{ point})$$

$$\begin{array}{r}
 0010 \quad 1000 \quad \cdot \quad 0000 \quad 0000 \\
 + 0000 \quad 0110 \quad \cdot \quad 0001 \quad 0010 \\
 \hline
 0010 \quad 1110 \quad \cdot \quad 0001 \quad 0010 \\
 \downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\
 3 \quad \quad 4 \quad \cdot \quad 1 \quad \quad 2
 \end{array}$$

(1 point)

$$28 + 6 \cdot 12 = 34 \cdot 12$$

d- $0.127 + 1.99$

$$\begin{array}{l}
 0 \cdot 1 \quad 2 \quad 7 \quad 1100 \\
 \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 0000 \quad 0001 \quad 0010 \quad 0111
 \end{array}$$

(1 point)

$$\begin{array}{r}
 0000 \quad \cdot \quad 0001 \quad 0010 \quad 0111 \\
 + 0001 \quad \cdot \quad 1001 \quad 1001 \quad 0000 \\
 \hline
 0010 \quad 1011 \quad 1011 \quad 0111 \\
 \downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\
 2 \quad \cdot \quad 1 \quad \quad 1 \quad \quad 7
 \end{array}$$

(1 point)

$$0.127 + 1.99 = 2.117$$

Q11) a) $(-6)_{10} : (000101)_2$ $(+5)_{10} : (010101)_2$

$$\begin{array}{r} 000101 \\ + 010101 \\ \hline \end{array}$$

$$+ 110101$$

$$\hline 111111 \rightarrow -1 \quad (0.5 \text{ pt})$$

no overflow (0.5 pt)

b) $(-1)_{10} : (111111)_2$ $(-7)_{10} : (111001)_2$

$$\begin{array}{r} 111111 \\ + 111001 \\ \hline \end{array}$$

$$+ 111001$$

$$\hline 1111000 \rightarrow 111000 \rightarrow -8 \quad (0.5 \text{ pt})$$

No overflow (0.5 pt)

c) $(+11)_{10} : (001011)_2$ $(+20)_{10} : (010100)_2$

$$\begin{array}{r} 001011 \\ + 010100 \\ \hline \end{array}$$

$$+ 010100$$

$$\hline 011111 \rightarrow +31 \quad (0.5 \text{ pt})$$

No overflow (0.5 pt)

d) $(-23)_{10} : (101001)_2$ $(-13)_{10} : (110011)_2$

$$\begin{array}{r} 101001 \\ + 110011 \\ \hline \end{array}$$

$$+ 110011$$

$$\hline \boxed{1}011100 \rightarrow 011100 \rightarrow +28 \quad (0.5 \text{ pt})$$

overflow since diff signs
of carry out bit & MSB
(0.5 pt)

Q12) a) 0111 1111 1111 (2 pt)

b) 1111 1111 1111 (2 pt)

c) 0111 1111 1111:-

Decimal: 2047 (1 pt)

Hex: 0111 → 7 1111 → F

7FF (1 pt)

1111 1111 1111:-

Decimal: 4095 (1 pt)

Hex: FFF (1 pt)

Q13)

1 pt each

Decimal	(0.5 pt)	(0.25 pt)	(0.25 pt)
	Signed Magnitude	Signed 1's Complement	Signed 2's Complement
-25	1001101	11100110	11100111
-64	11000000	10111111	11000000
-32	10100000	11011111	11100000
+45	00101101	00101101	00101101
+127	01111111	01111111	01111111
+0	00000000	00000000	00000000

Q14) I.) count of 1's: 5

even parity bit: 1 (1 pt)

II.) count of 1's: 4

even parity bit: 0 (1 pt)

Q15 a) Add 3 to each quadruplet (1 pt)

$$\begin{array}{r} \underline{0110} \\ 6+3=9 \end{array} \quad \begin{array}{r} \underline{1001} \\ 9+3=11 \end{array} \quad \begin{array}{r} \underline{0111} \\ 7+3=10 \end{array} \quad \begin{array}{r} \underline{0011} \\ 3+3=6 \end{array}$$

(2 pt)

1001 1100 1010 0110 Ans. (1 pt)

b) MSB will be same then XOR all bits (1 pt)

010010011110101

	XOR
0 → 0	0 0 0
0 ⊗ 1 → 1	0 1 1
1 ⊗ 0 → 1	1 0 1
0 ⊗ 0 → 0	1 1 0
0 ⊗ 1 → 1	
1 ⊗ 0 → 1	
0 ⊗ 0 → 0	
0 ⊗ 1 → 1	
1 ⊗ 1 → 0	
1 ⊗ 1 → 0	
1 ⊗ 1 → 0	
1 ⊗ 1 → 0	
1 ⊗ 0 → 1	

(2 pt for correct working)

$$0 \oplus 1 \rightarrow 1$$

$$1 \oplus 0 \rightarrow 1$$

$$0 \oplus 1 \rightarrow 1$$

Gray code: 010110100001111 (1 pt)